SWERVE DRIVE SPECIALTIES

MK5n Swerve Module Assembly Guide

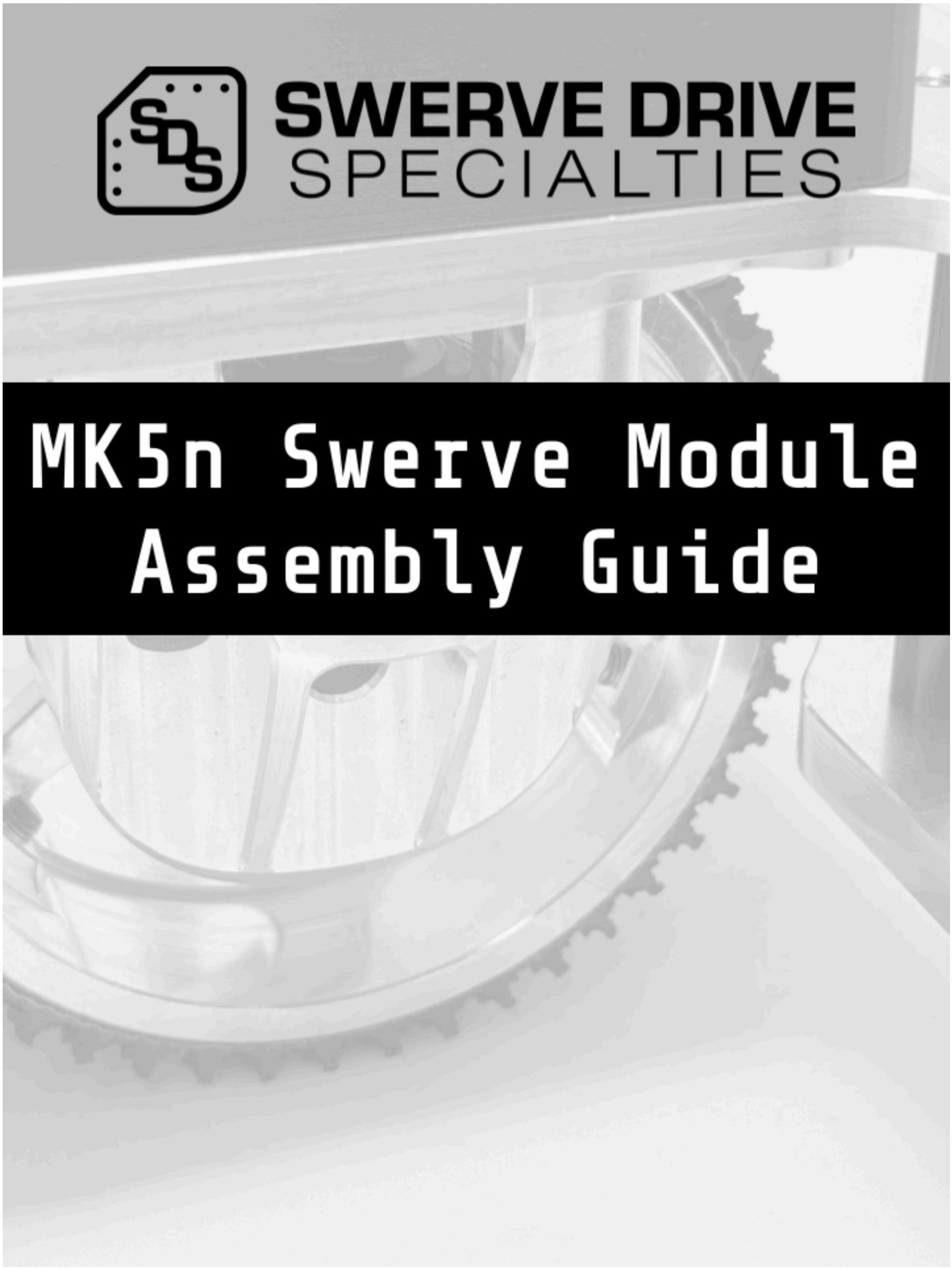


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Overview

Swerve Module Configurations

The swerve module can be configured as follows:

- Drive Motor Type: Kraken X60, NEO or NEO Vortex
- Steering Motor Type: Kraken X44
- Wheel Type: Molded Spike Grip or Billet with Neoprene
- Layout: A or B

Please note:

- *Drive and Steering Motor Types* refer to the motor-specific components provided in the kit, such as the pinion gears. The motors are not included.
- The swerve module kit comes with the configurations selected when ordered. The kit does not include all of the components for all of the configurations.

Assembly Guide Conventions

The following configuration is primarily used for illustration purposes:

- Drive Motor Type: Kraken X60
- Steering Motor Type: Kraken X44
- Gear Ratio: R3
- Wheel Type: Molded Spike Grip
- Layout: B

Details of particular importance are indicated with **IMPORTANT**. Not following them could significantly impact the operation of the swerve module.

Supplies needed

Tools

- T25 L-key or T-handle (for motor shaft screws)
- 1/8” hex L-key or T-handle (for #10 button head screws)
- 5/32” hex T-Handle (for #10 socket head screws)
- 3/8” open-end wrench (for holding the Kraken pinion gears)
- 1/2” open-end wrench (for holding the NEO or NEO Vortex drive pinion gears)
- #1 phillips screwdriver (for encoder cover and encoder housing)

Overview, cont'd

Threadlocker and Retaining Compound

- Henkel Loctite 243 Threadlocker or equivalent (for screws)
- Henkel Loctite 609 Retaining Compound or equivalent (for encoder magnet)

Use of Threadlocker

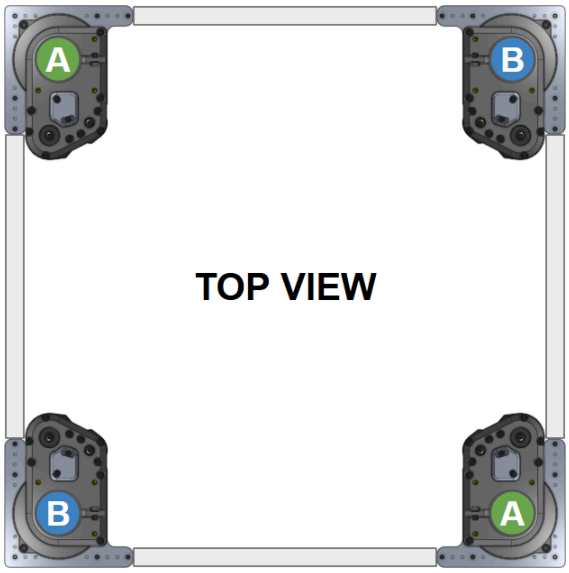
IMPORTANT Use threadlocker to secure all screws to ensure they do not inadvertently come loose, except the screws used to secure the steering encoder, as the threadlocker may adversely react with the encoder housing. The Encoder Cover screws also do not require threadlocker.

Lubrication

IMPORTANT Lubricate all gear teeth with Lucas Red "N" Tacky grease before using the swerve module and regularly with use, or the gears may wear out prematurely.

Swerve Module Layout

The MK5n Swerve Module does not have a symmetric footprint, so it is available in mirrored layouts known as "A" and "B". A typical drive base will have two A layouts and two B layouts:



The assembly process for layout A and B is the same, except:

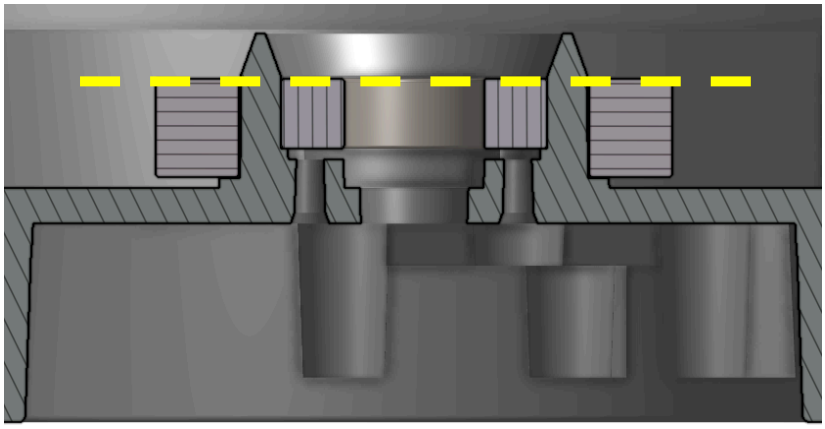
- The arrangement of components is generally mirrored.
- The following components differ between layout A and B, and are marked "A" or "B", respectively:
    - Main Plate
    - Shift Plate
    - Housing

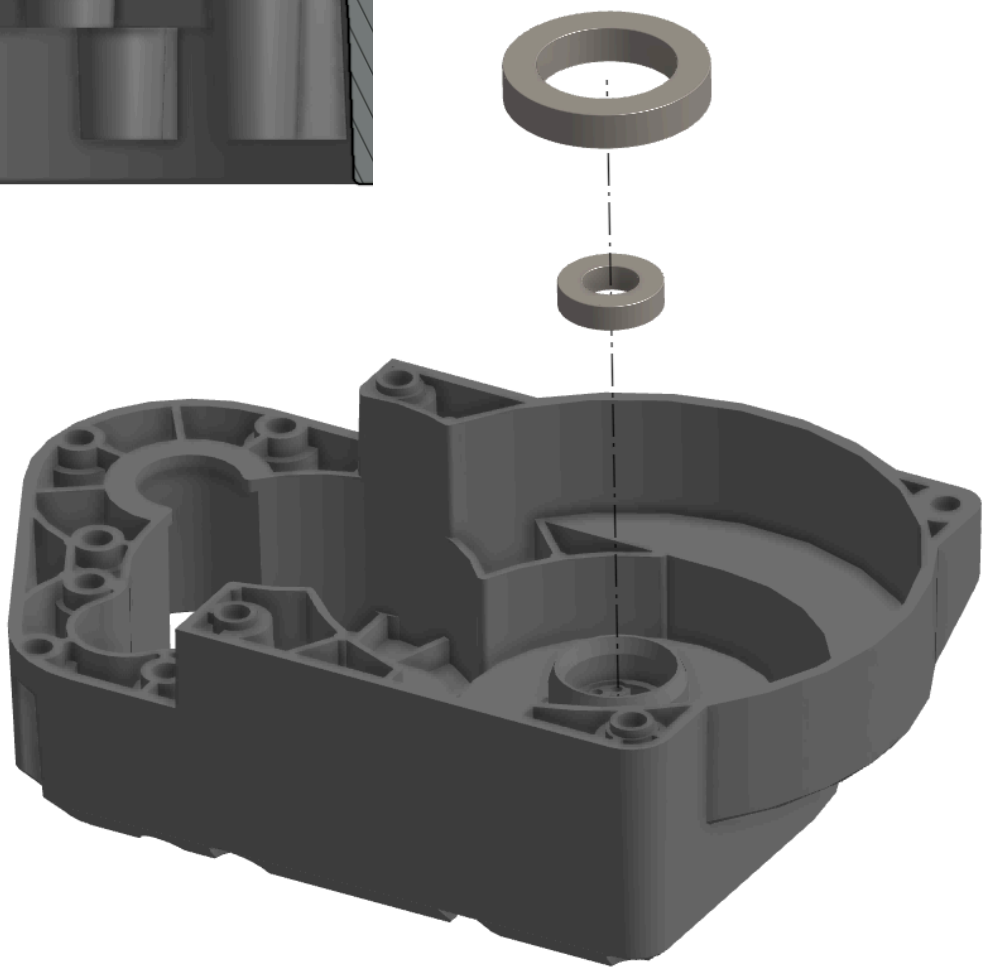
Step 1: Housing Bearing Installation

Install the 6800ZZ and 6805ZZ bearing into the Housing. To help fully seat the bearings:

- The post on the Main Steering Gear (shown in Step 5) can be used to press the 6800ZZ bearing in.
- The bearing pocket in the larger end of the Drive Double Gear (shown in Step 20) can be used to press the 6805ZZ bearing in.

IMPORTANT Ensure all both bearings are fully seated. The top surfaces (as shown) of the 6800ZZ and 6805ZZ bearings should be flush with the base of the housing's bearing pocket lead-in chamfers.

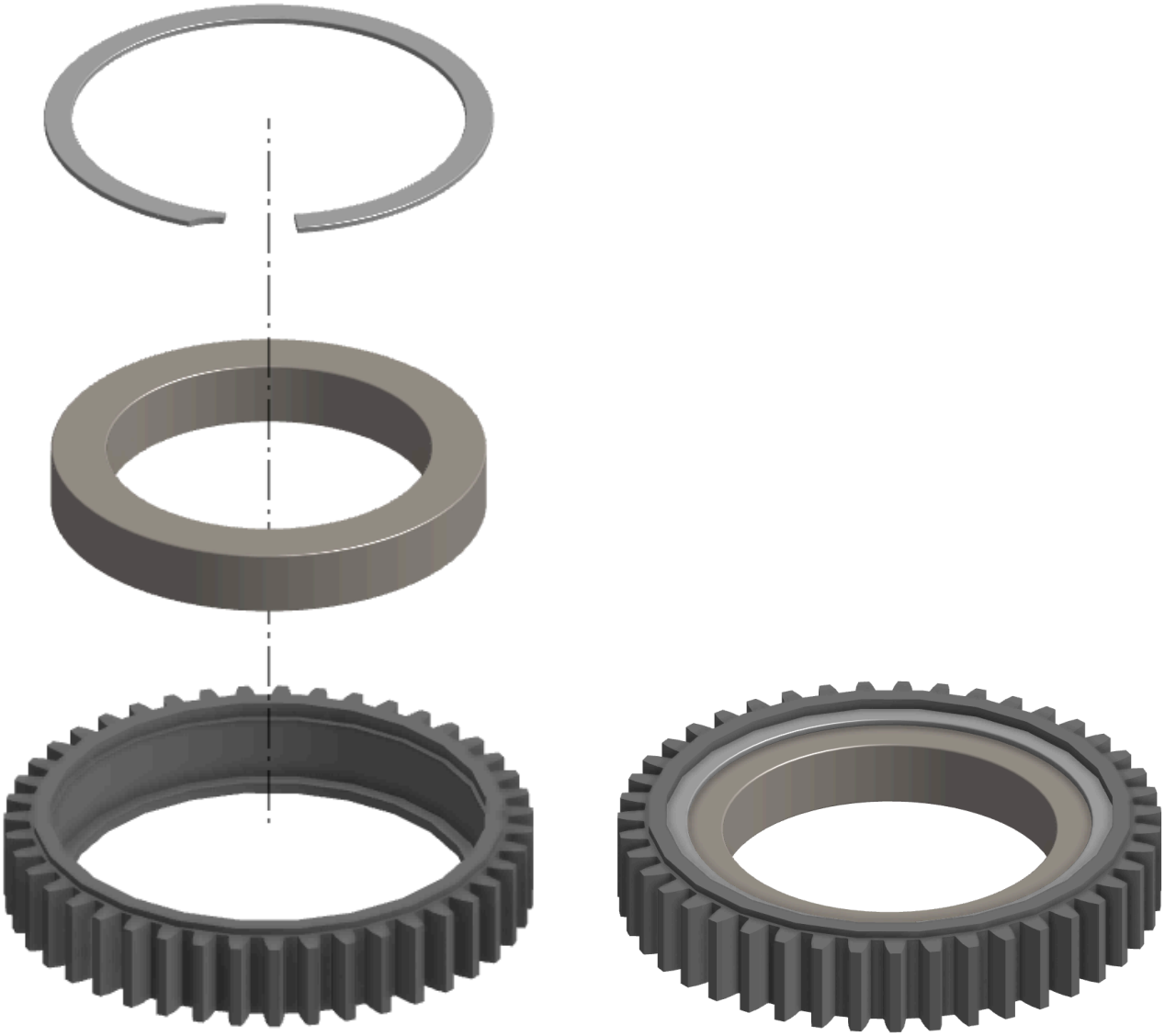




Step 2: Idler Gear Subassembly

Insert the 6807ZZ bearing into the Idler Gear, and then secure it by snapping the retaining ring into the groove.

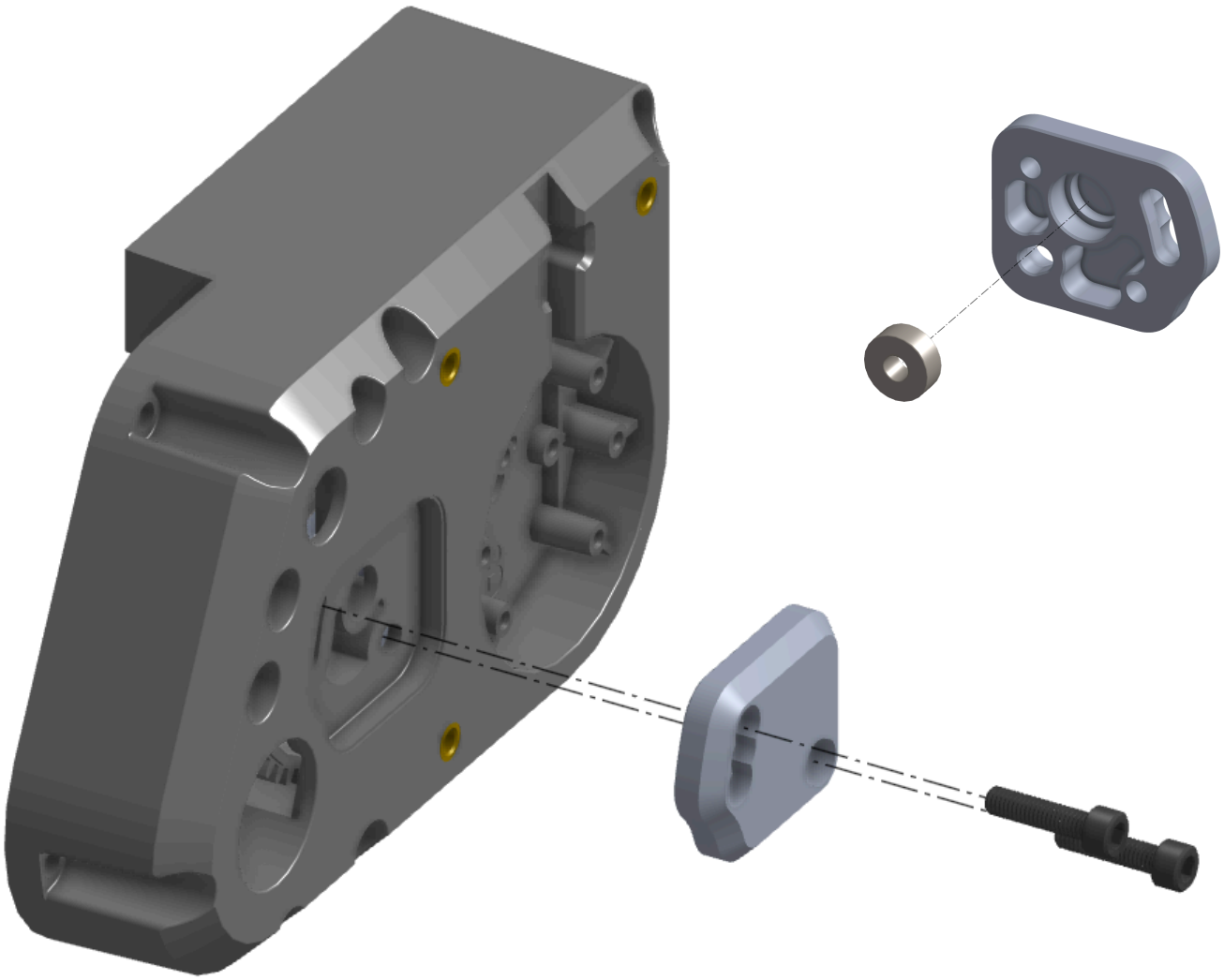
IMPORTANT Ensure the retaining ring is fully seated.



Step 3: Shift Plate Installation

Insert an R3ZZ bearing into the Shift Plate, the Shift Plate into the Housing, and then two 10-32 x 3/4” socket screws through the Shift Plate.

IMPORTANT Ensure the screw in the triple-pocket slot is in the pocket marked with the same drive gear ratio number of the Drive Pinion Gear that will be used (1, 2, or 3 for R1, R2, or R3, respectively).

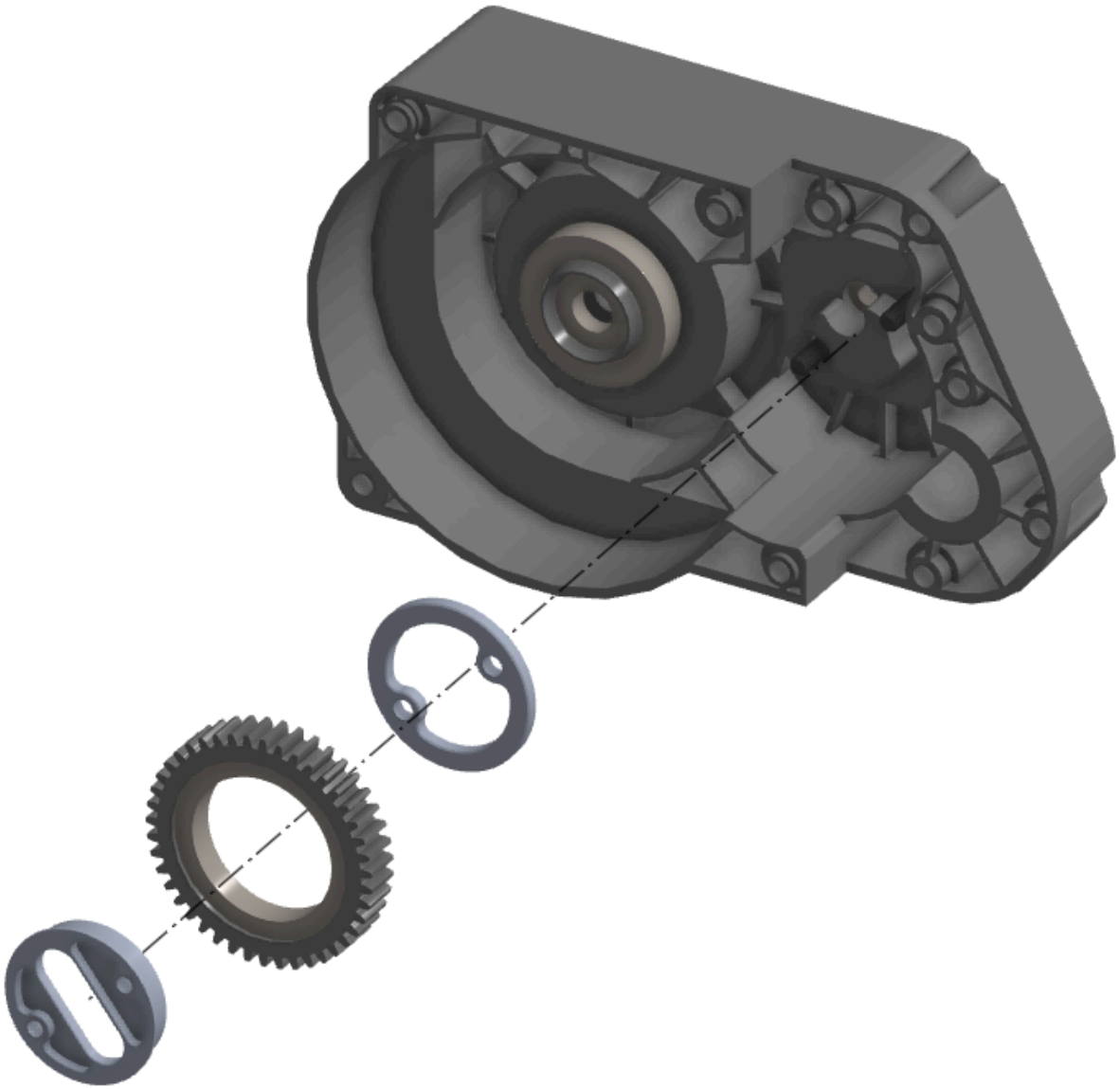


Step 4: Idler Gear Installation

Place the Idler Spacer over the Shift Plate screws, insert the Idler Gear Retainer into the Idler Gear subassembly, and then secure the gear and retainer with the previously installed screws. The Idler Gear can be installed in either orientation (retaining ring facing up or down).

IMPORTANT Ensure the screw in the triple-pocket slot is in the pocket labeled with the same drive gear ratio number of the Drive Pinion Gear that will be used (1, 2, or 3 for R1, R2, or R3, respectively).

IMPORTANT Ensure the R3ZZ bearing does not fall out of the Shift Plate while securing the Idler Gear components.



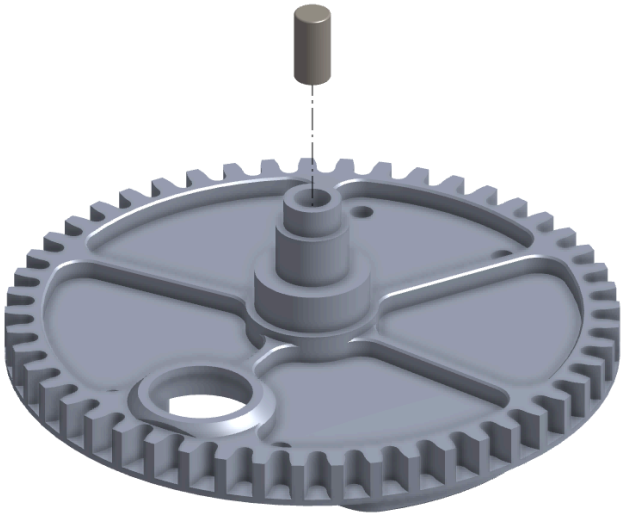
Step 5: Encoder Magnet Installation

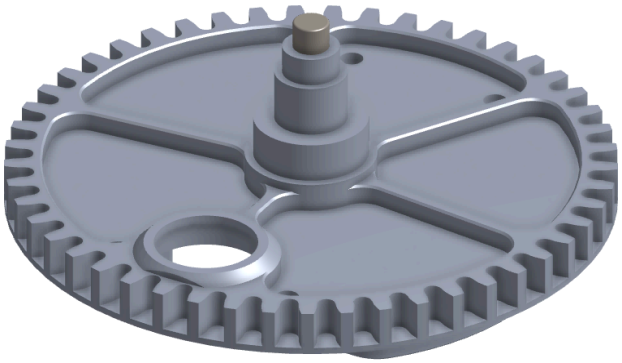
Bond the magnet to the Main Steering Gear:

1. Apply Loctite 609 retaining compound (or equivalent) to the magnet.
2. Insert the magnet into the 0.25” hole in the Center Column.
3. Wipe off any excess retaining compound from the magnet and Main Steering Gear.
4. Before continuing assembly, set this subassembly aside with the magnet facing upward so the retaining compound can cure without the magnet inadvertently shifting.

Please note the magnet is included with the encoder, and not with the swerve module kit.

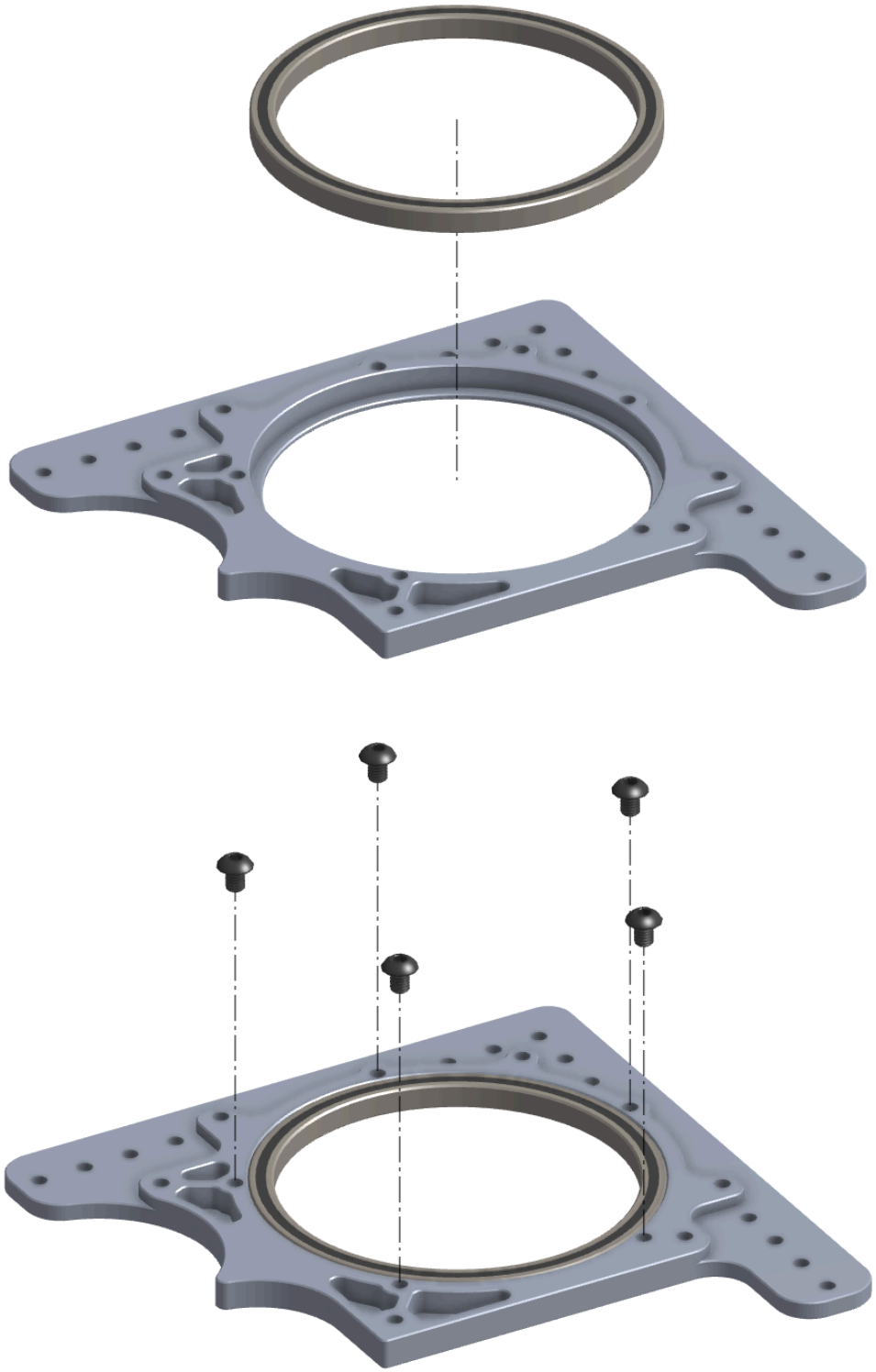
IMPORTANT Ensure the magnet is fully seated and subsequently cured before continuing assembly.





Step 6: Steering Bearing Installation

Install the 4” OD Main Steering Bearing into the Main Plate, and then secure it with five #10-32 x 0.25” button screws.



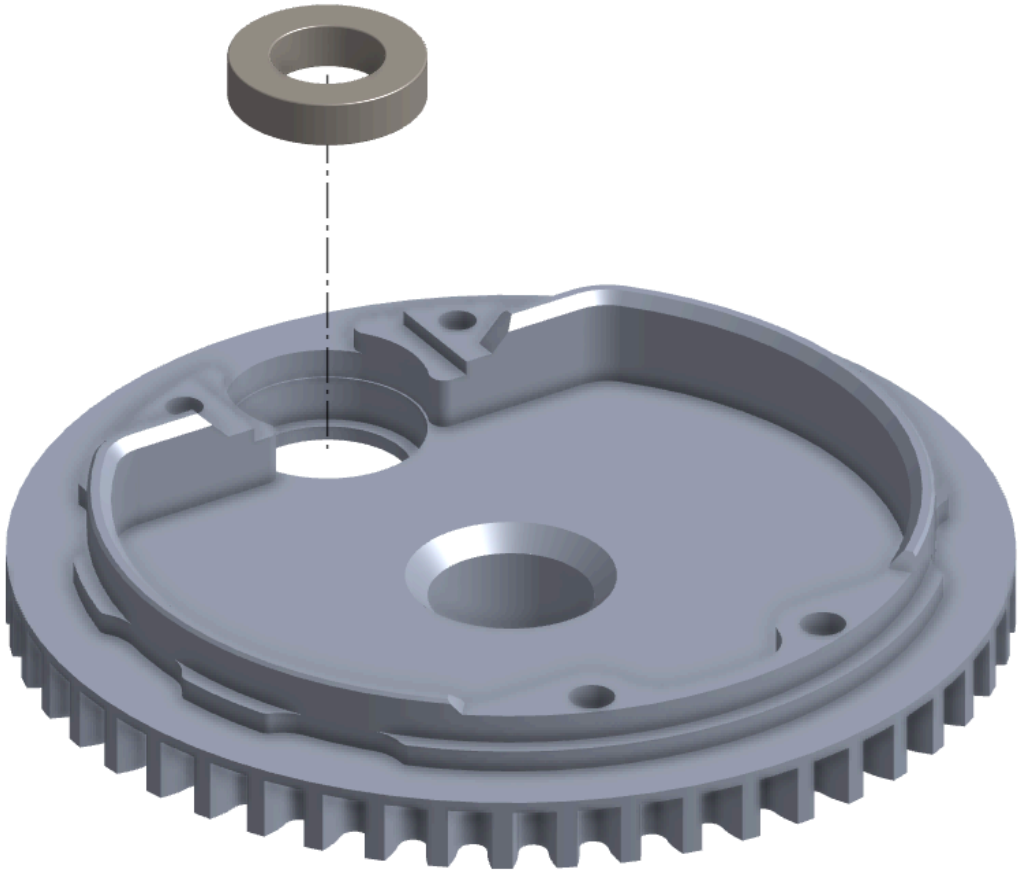
Step 7: Bottom Plate Installation

Attach the Bottom Plate to the Main Plate with a 10-32 x 2.5” socket screw and Spacer. Attaching it at this time is not required, though it eases the assembly process by enabling the swerve module to rest on its side.



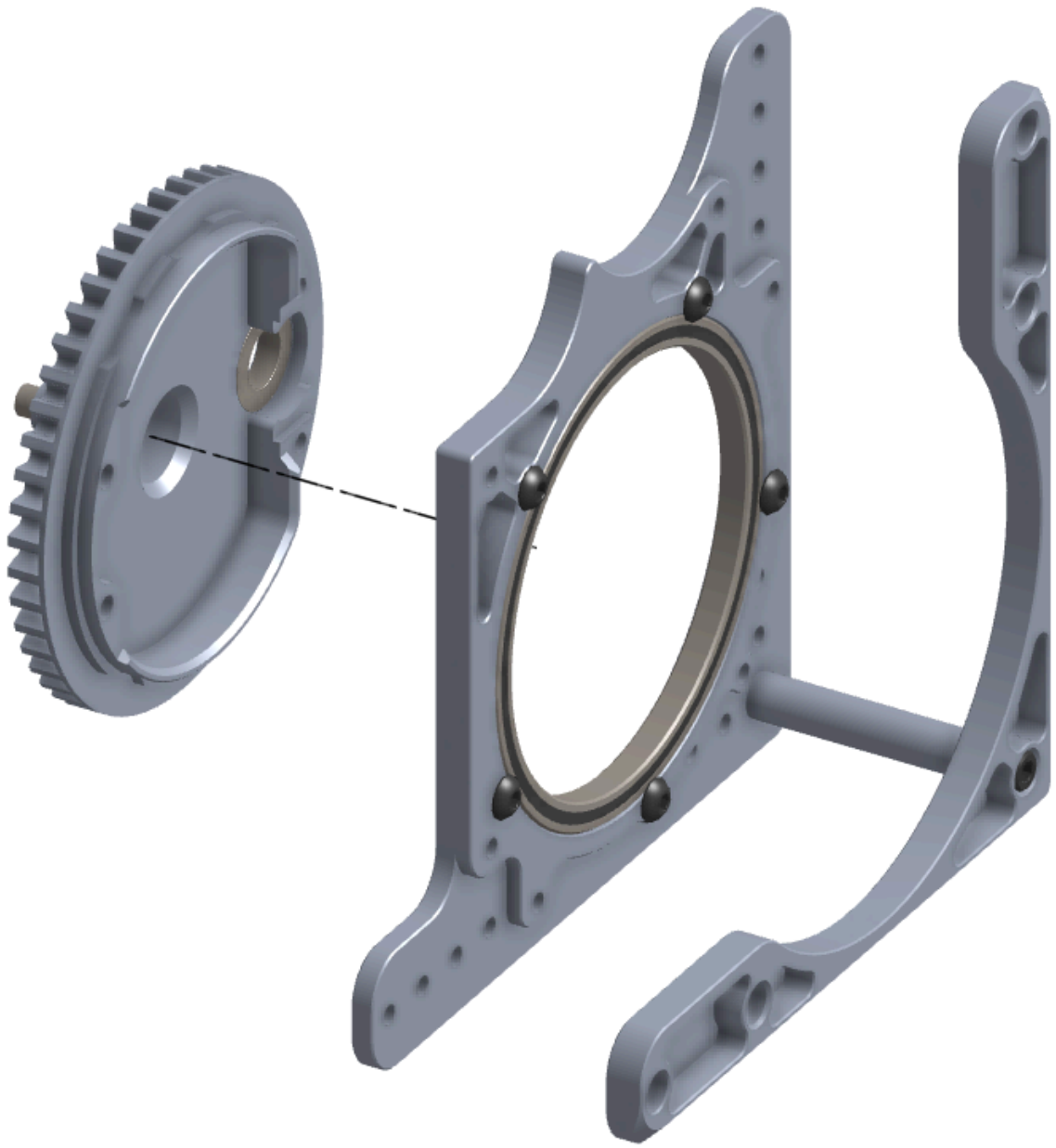
Step 8: Main Steering Gear Subassembly

Insert the 6801ZZ bearing into the Main Steering Gear.



Step 9: Main Steering Gear Installation

Insert the Main Steering Gear subassembly into the Main Steering Bearing.



Step 10: 15T Bevel Gear Subassembly

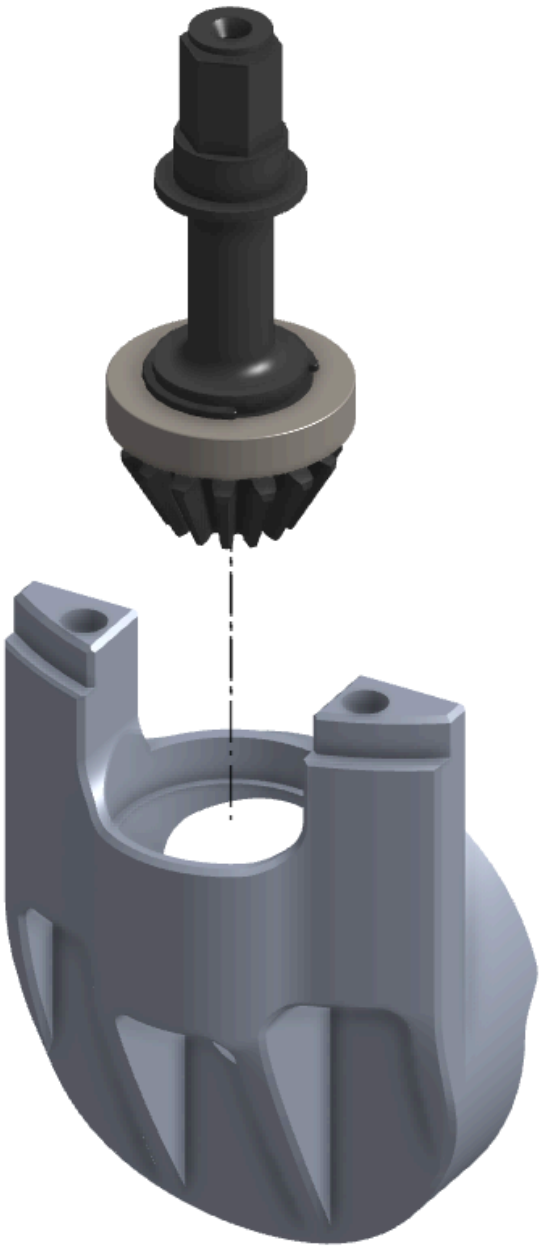
Place the 6803ZZ bearing onto the 15T Bevel Gear, and then secure it by snapping the crescent ring into the groove.

IMPORTANT Ensure the crescent ring is fully seated in the groove.



Step 11: 15T Bevel Gear Installation

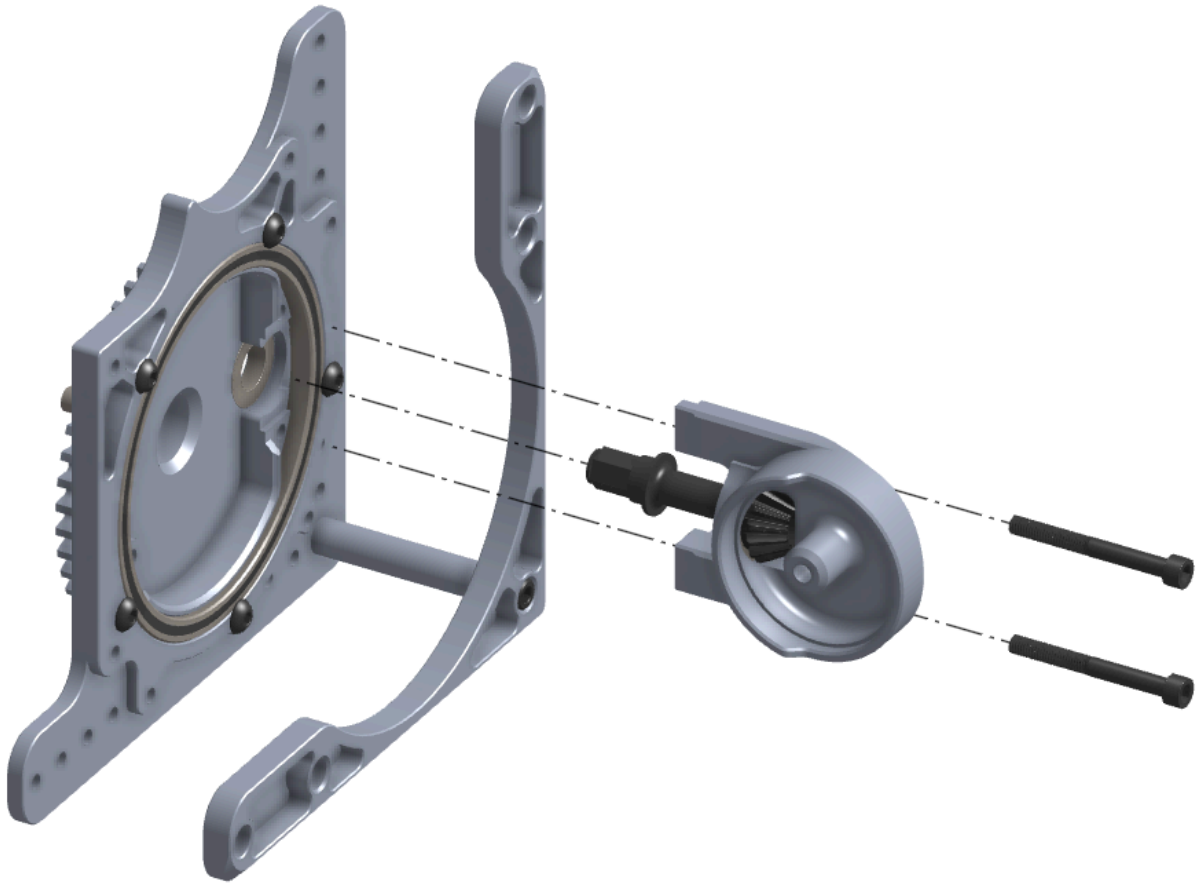
Insert the 15T Bevel Gear subassembly into Wheel Mount B. The bearing should fully seat into the wheel mount.



Step 12: Wheel Mount B Installation

Place Wheel Mount B onto the Steering Bearing and Main Steering Gear by passing the 15T Gear through the 6801ZZ bearing, and then secure it with two 10-32 x 1.75” socket screws.

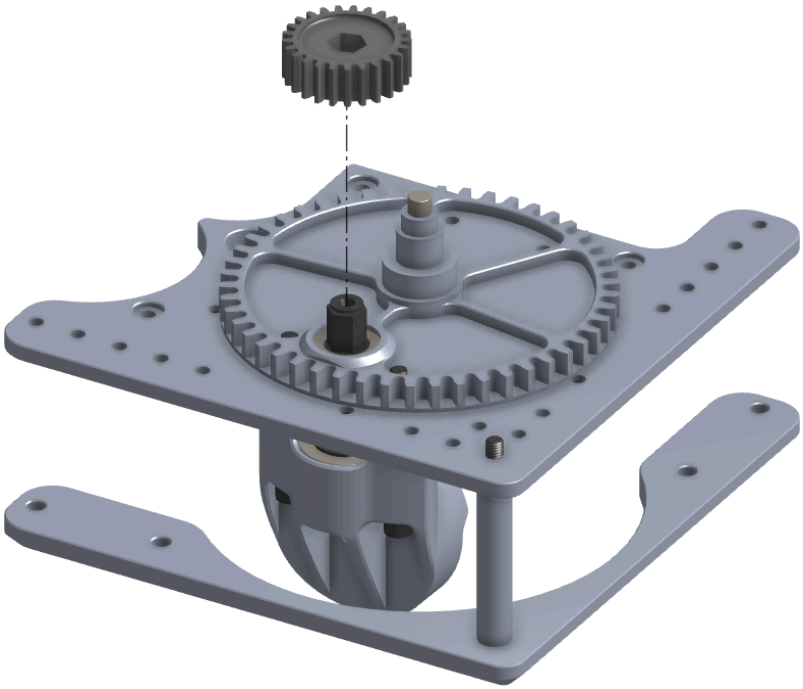
IMPORTANT Ensure the wheel mount is fully seated and contacting the Steering Bearing’s inner race before tightening the screws.

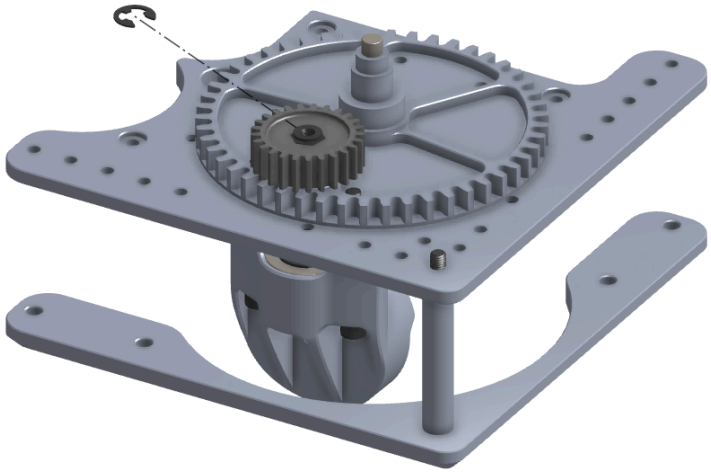


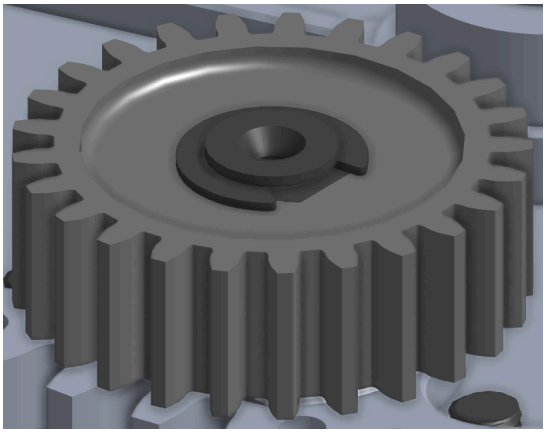
Step 13: Intermediate Gear Installation

Place the Intermediate Gear pocketed side facing down onto the 15T Bevel Gear, and then secure it by snapping the e-clip into the groove.

IMPORTANT Ensure the e-clip is fully seated in the groove.







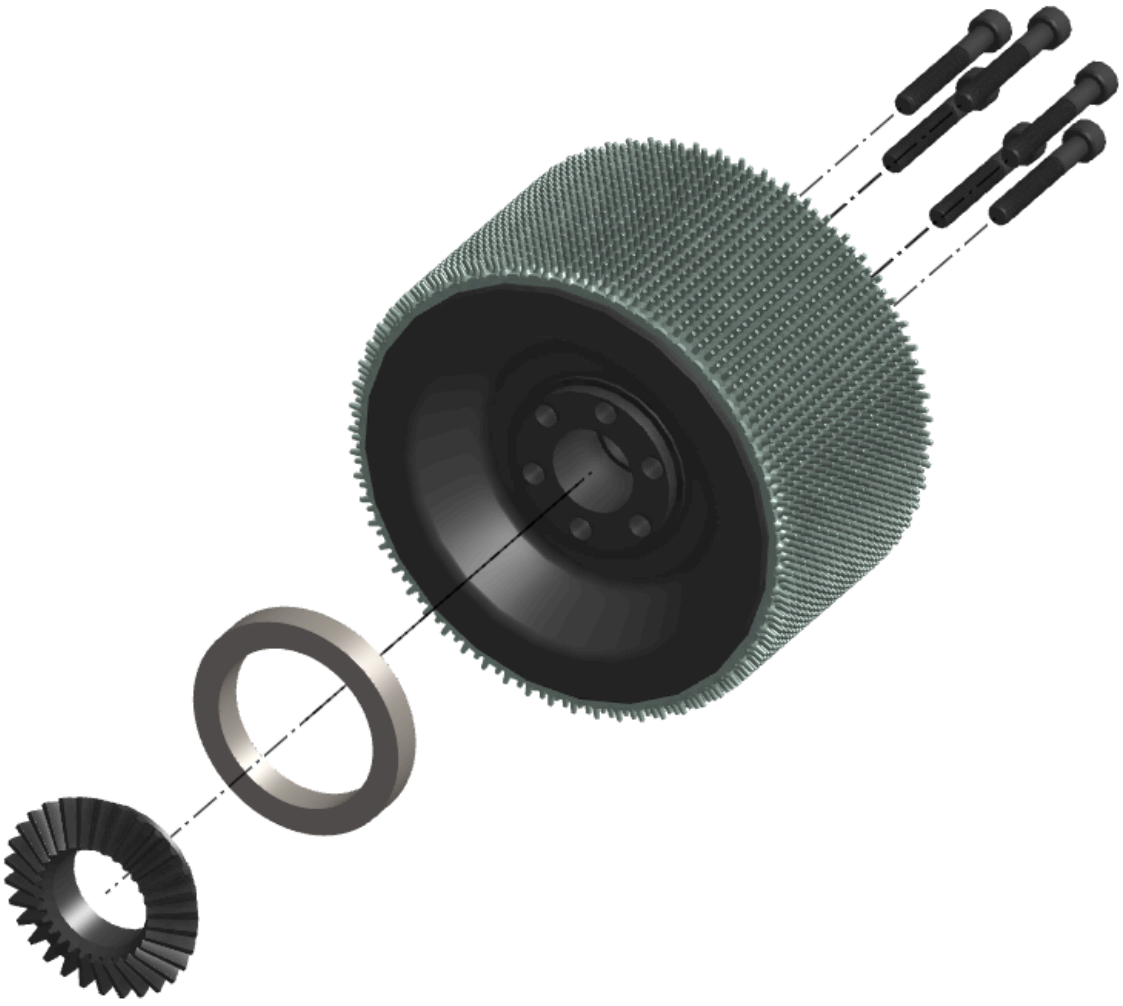
Step 14: Wheel Subassembly

Install the 6807ZZ bearing onto the wheel, the 30T Bevel Gear into the 6807ZZ bearing, and then secure them with:

- Billet Wheel: Six 10-32 x 3/4” socket screws
- Molded Wheel: Six 10-32 x 1.25” socket screws

Tighten the screws to a torque of 2.5 Nm.

IMPORTANT Ensure the bearing and bevel gear are fully seated before tightening the screws.

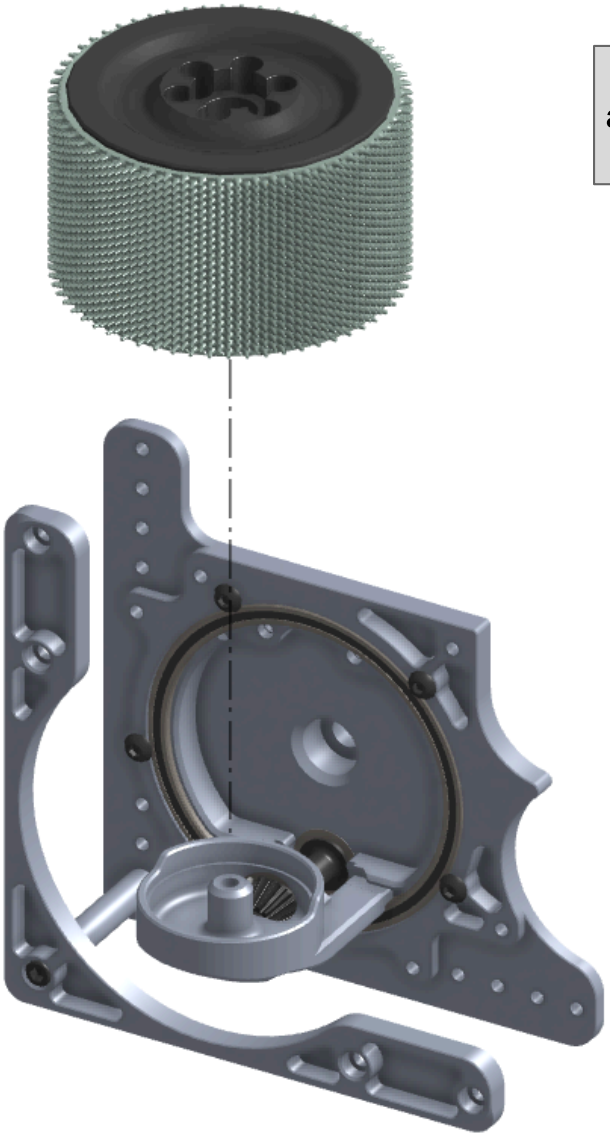


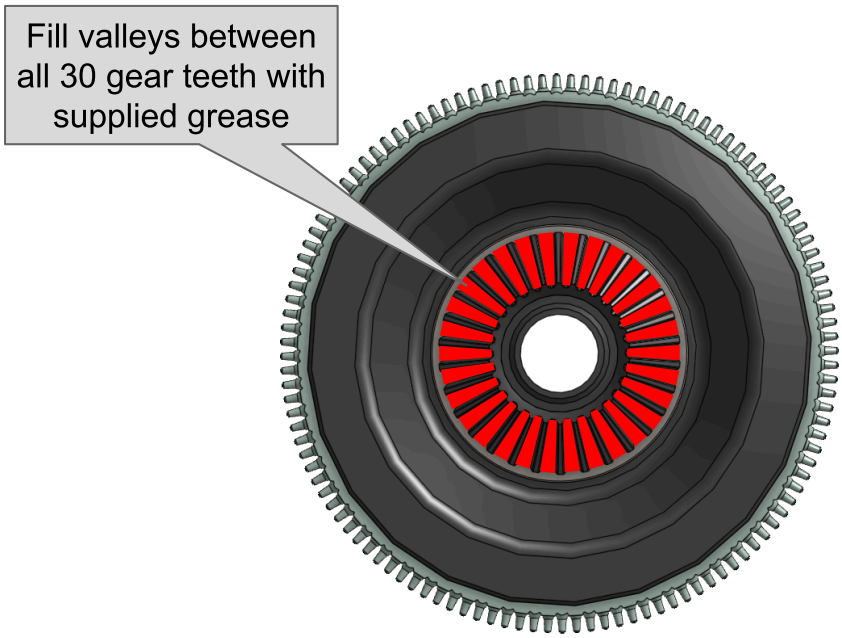
Step 15: Wheel Installation

Grease the 30T Bevel Gear teeth with part of the included grease packet by filling the valleys between all 30 teeth, and then install the wheel into Wheel Mount B.

Most of the grease should still be in the packet for greasing the remaining gears in Step 21.

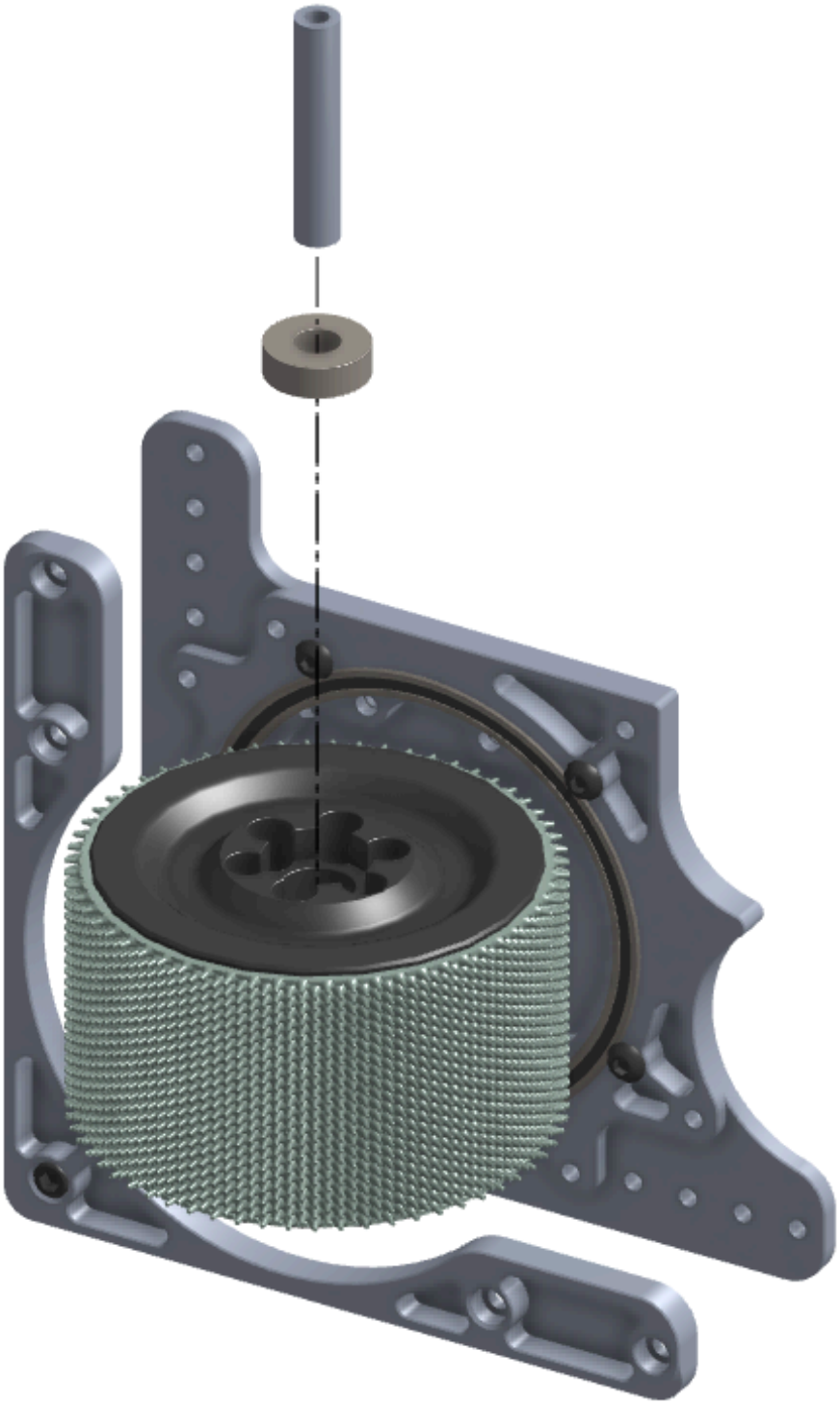
IMPORTANT Ensure the 30T Bevel Gear is greased before continuing with assembly.





Step 16: Wheel Installation, cont’d

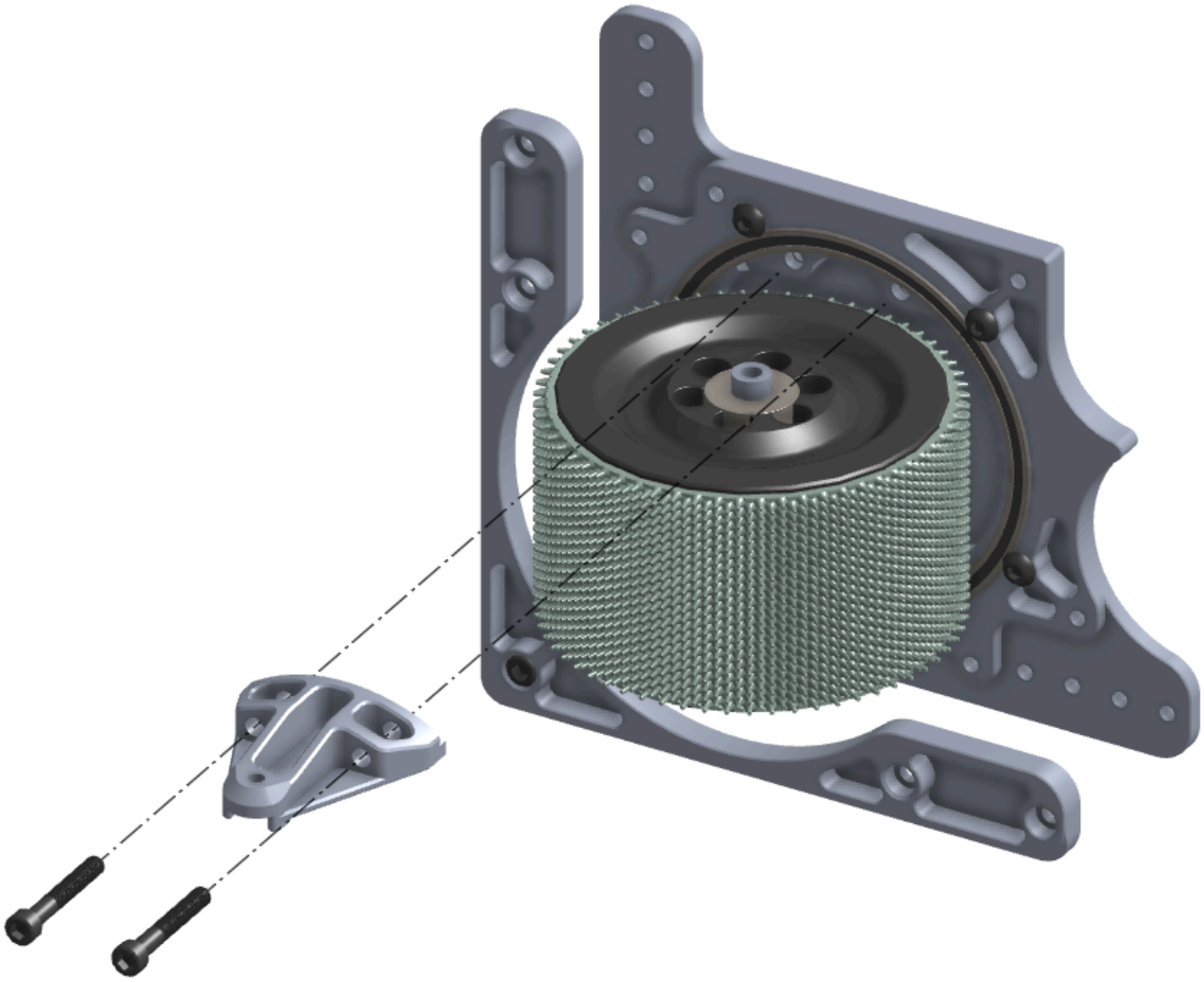
Insert the R6ZZ bearing into the wheel, and then the Spacer into the bearing.



Step 17: Wheel Mount A Installation

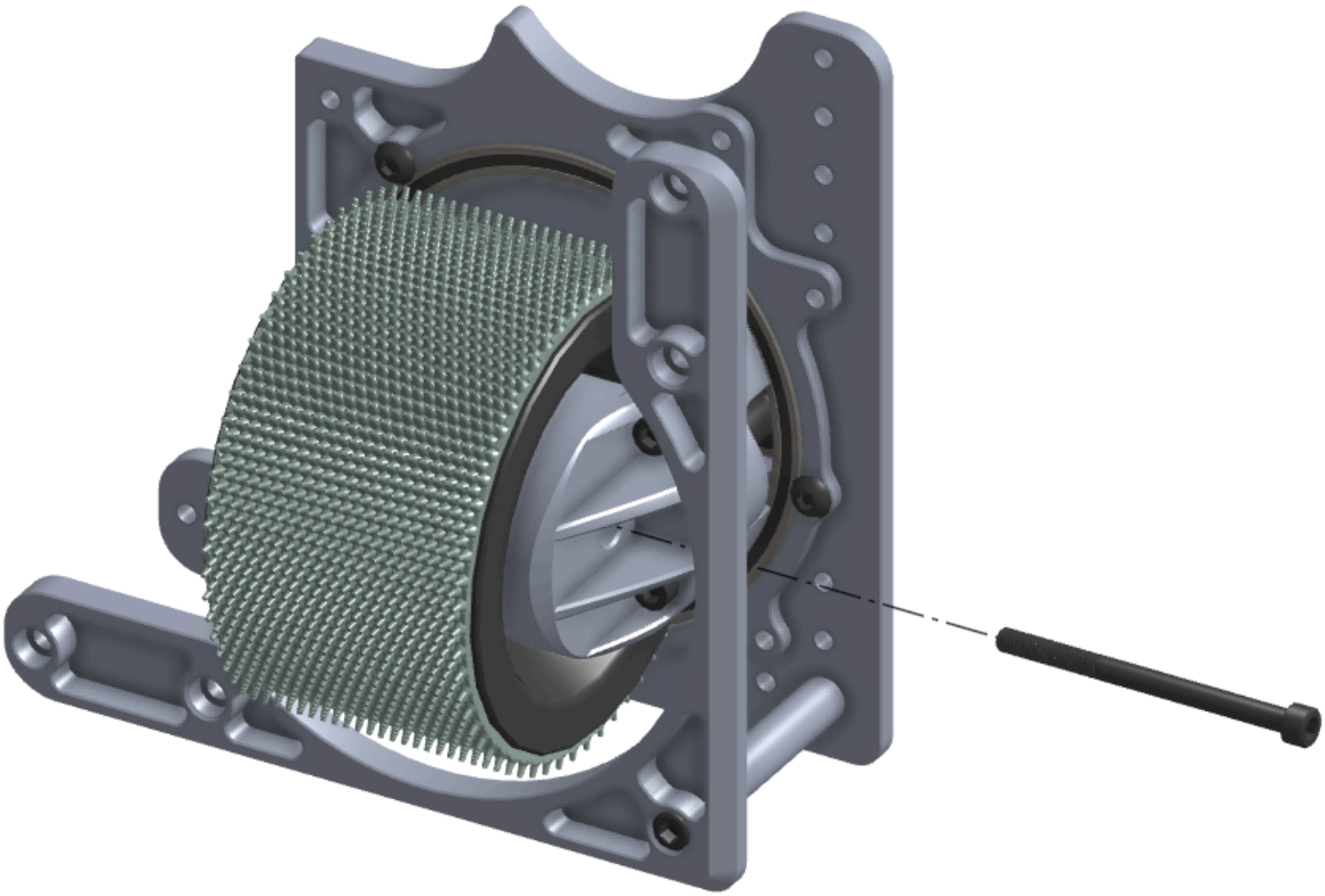
Install Wheel Mount A. It goes in at a slight angle and then gets tilted into place (the wheel also may need to be tilted slightly during the installation process), and then secure it with two 10-32 x 1.25” socket screws.

IMPORTANT Ensure the wheel mount is fully seated and contacting the Steering Bearing’s inner race before tightening the screws.



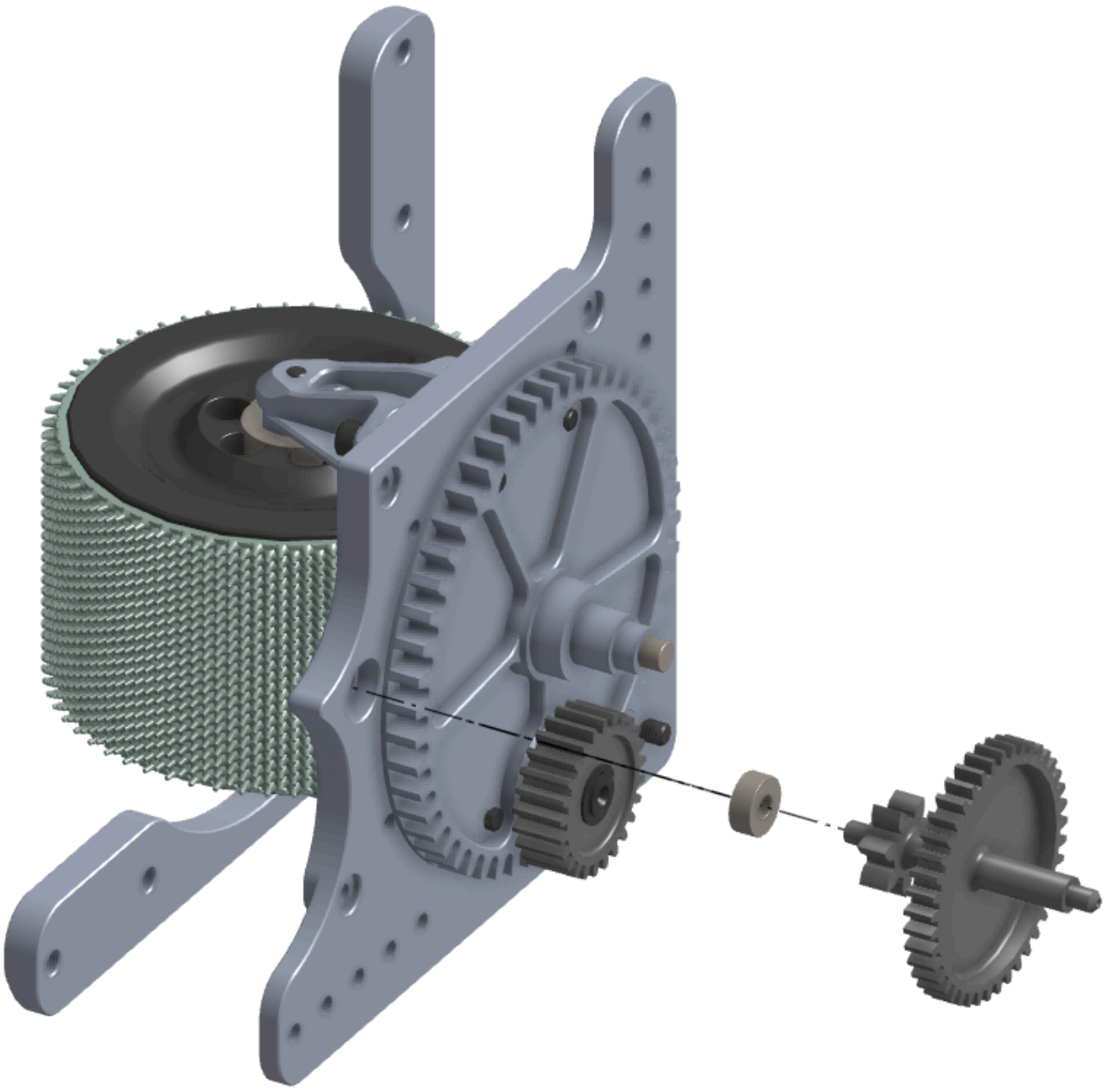
Step 18: Wheel Axle Screw Installation

Secure the wheel with a 10-32 x 2.5” socket screw.



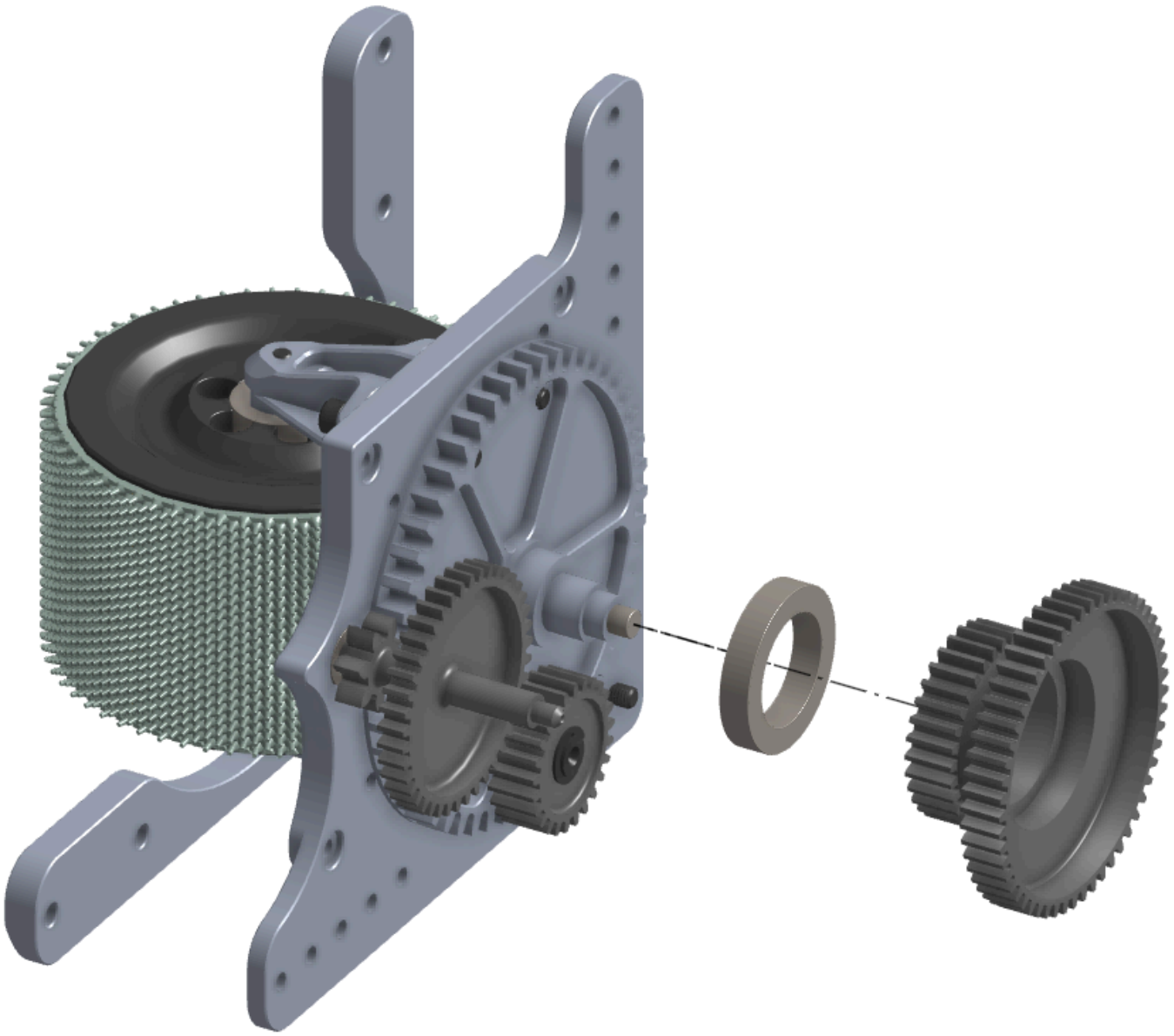
Step 19: Steering Double Gear Installation

Insert an R3ZZ bearing into the Main Plate, and then the Steering Double Gear into the R3ZZ bearing.



Step 20: Drive Double Gear Installation

Place the 6804ZZ bearing onto the Main Steering Gear, and then the Drive Double Gear onto the 6804ZZ bearing.



Step 21: Housing Installation

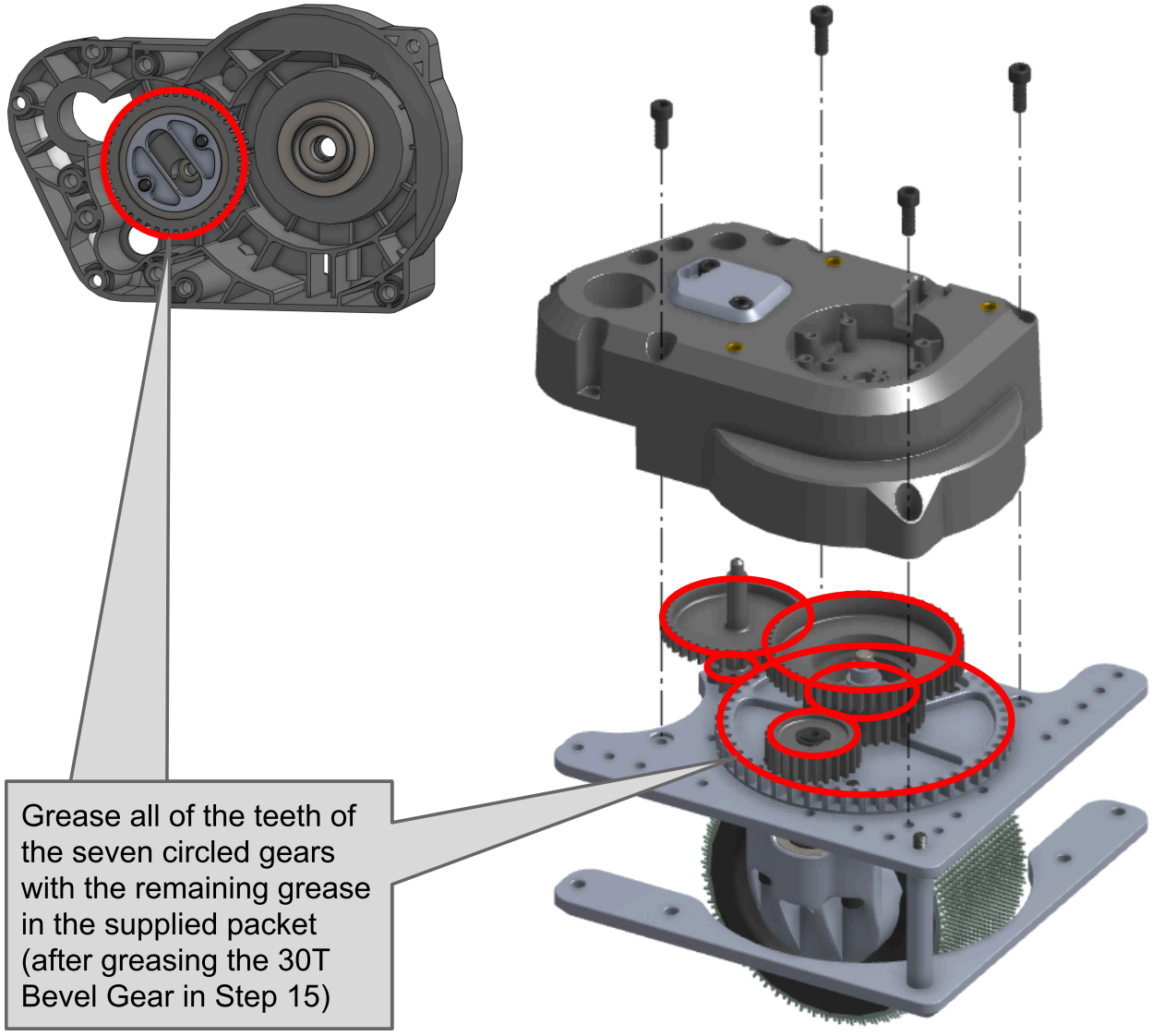
Grease all of the gear teeth:

- Idler Gear (attached to the Housing)
- Main Steering Gear
- Lower and upper gears of the Steering Double Gear
- Intermediate Gear
- Lower and upper gears of Drive Double Gear.

All of the grease should now be consumed from the grease packet.

Install the Housing onto the Main Plate, and then secure it with four 10-32 x 1/2” socket screws.

IMPORTANT Ensure all of the gears are greased before continuing with assembly.

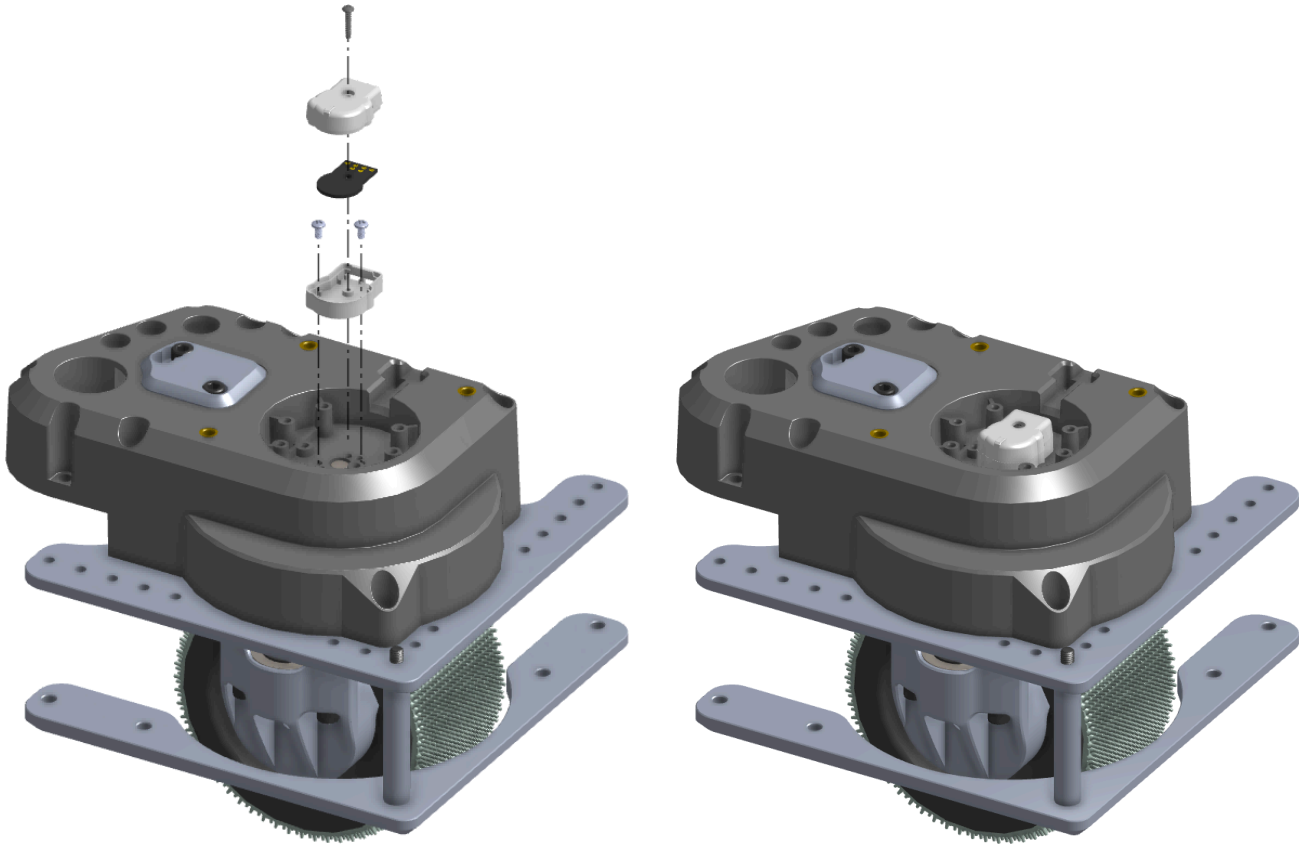


Step 22: Encoder Installation

Install the encoder with the encoder-supplied screws.

IMPORTANT Do not overtighten the screws, and do not use threadlocker.

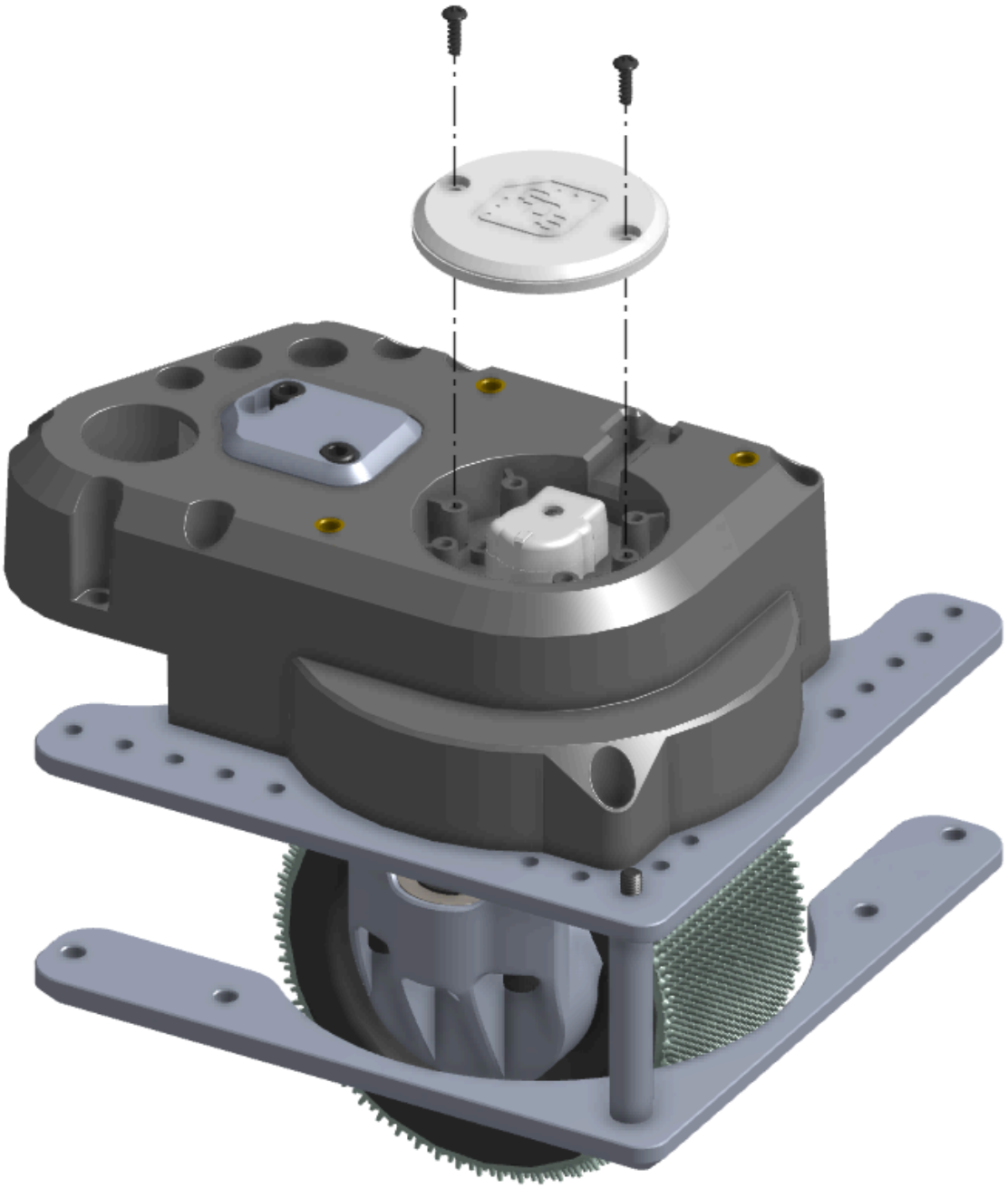
Secure the encoder wires in the groove of the Housing with a cable tie going through the Housing. Please note one of the cable tie pass-through openings is larger to accommodate the cable tie's head.



Step 23: Encoder Cover Installation

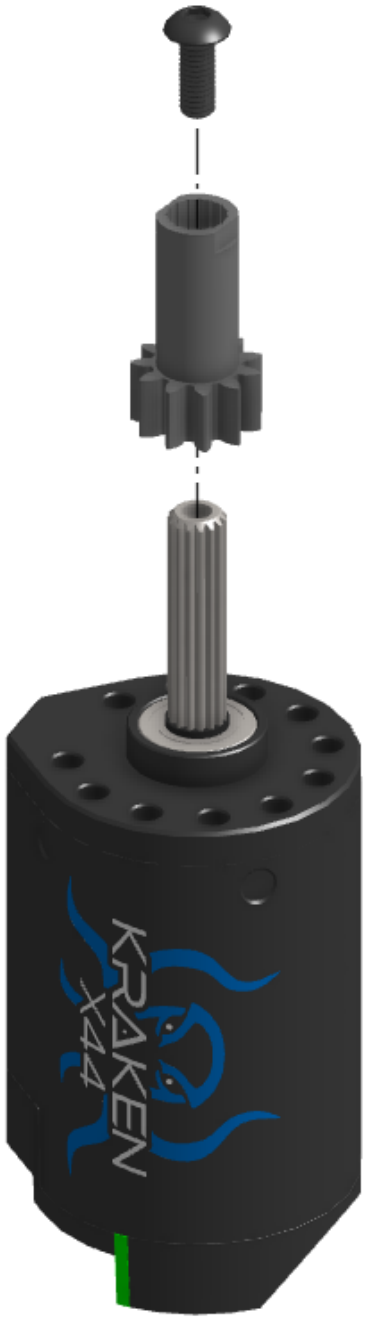
Install the Encoder Cover, and then secure it with two 4-20 x 3/8” thread-forming screws.

IMPORTANT Do not overtighten the screws, and do not use threadlocker.



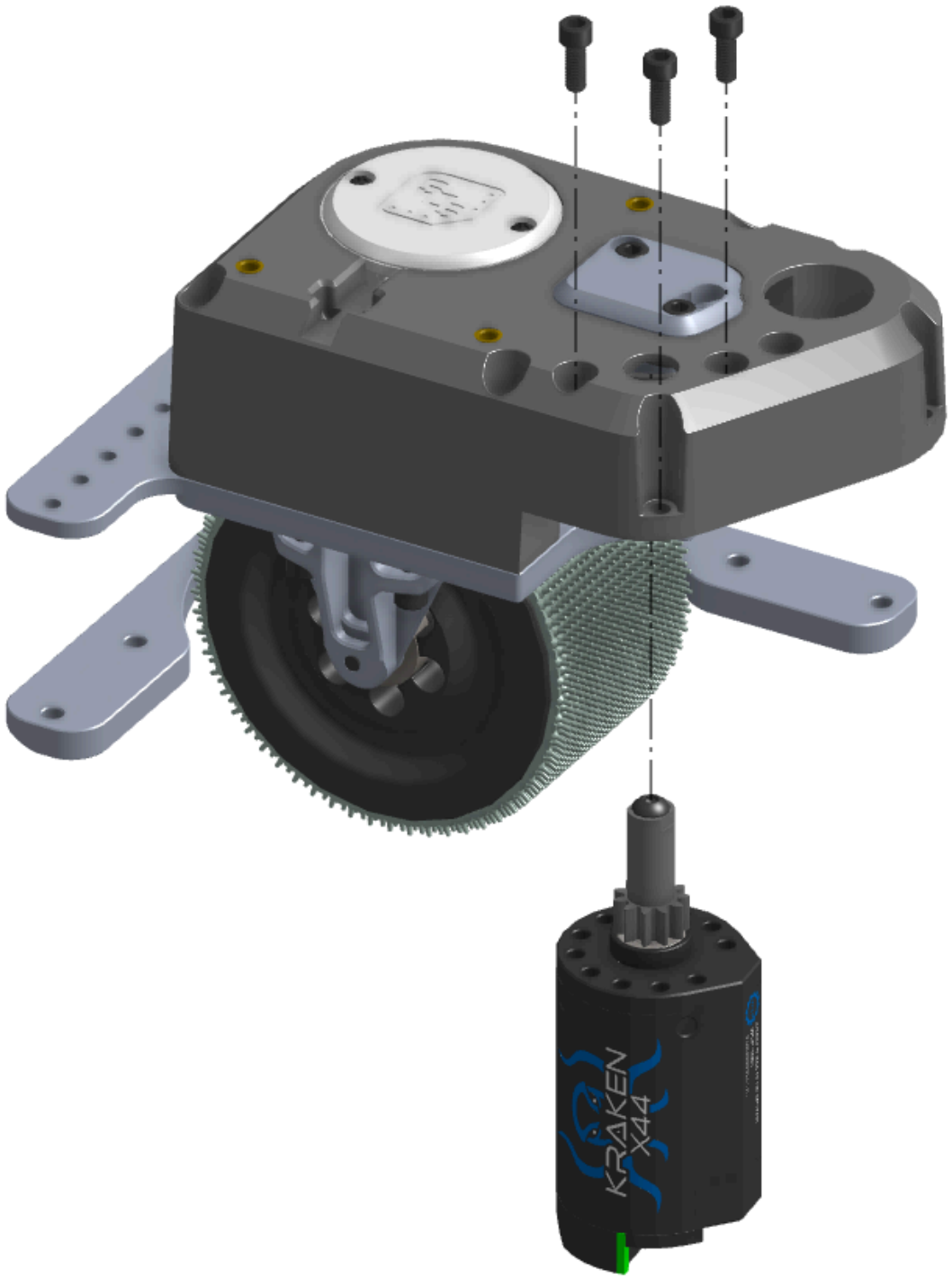
Step 24: Steering Pinion Gear Installation

Place the Steering Pinion Gear onto the Kraken X44 motor shaft, and then secure it with a 10-32 x 3/8" button screw (screw drive is T25, not 1/8” hex). Use a 3/8” open-end wrench on the flats at the top of the gear to prevent it from rotating while tightening the screw.



Step 25: Steering Motor Installation

Install the steering motor into the Housing, and then secure it with three 10-32 x 1/2” socket screws.

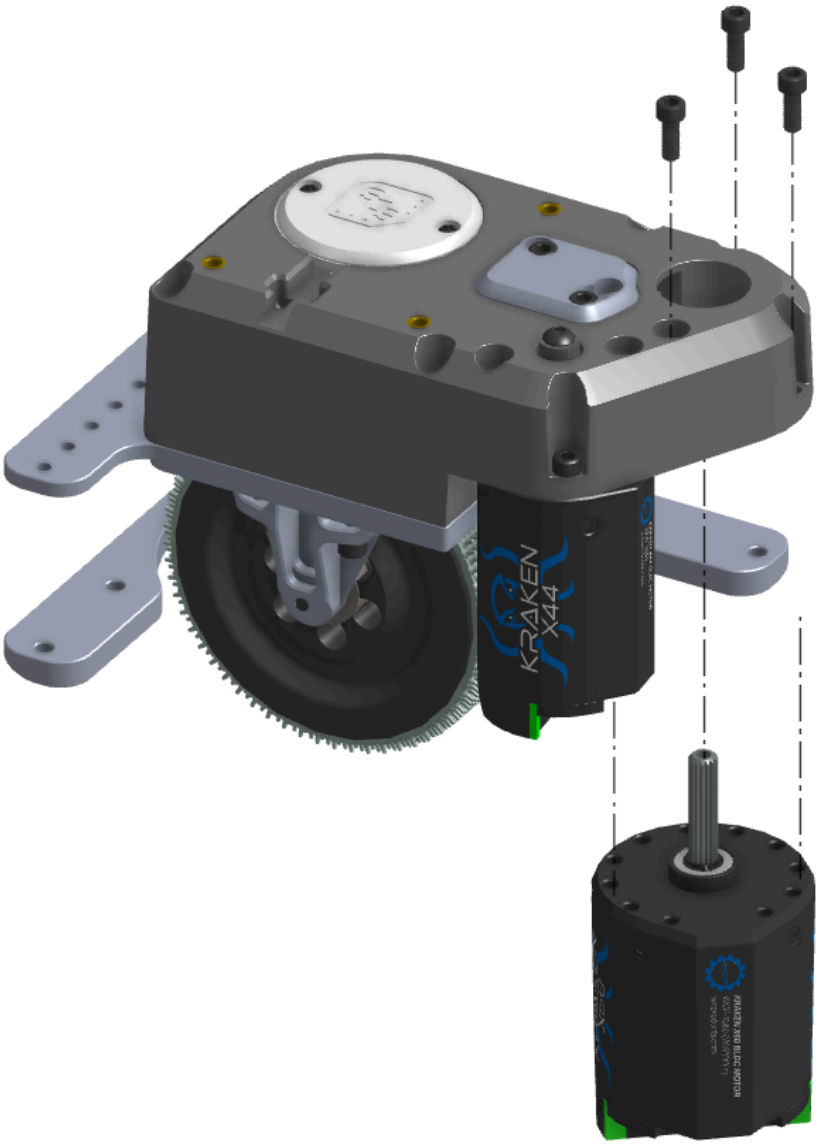


Step 26: Drive Motor Installation

If using a Kraken X60 drive motor, remove the motor's cooling port screw closest to the shaft and cover that port with a piece of tape.

Install the drive motor into the Housing, and then secure it with three 10-32 x 1/2" socket screws.

IMPORTANT For the Kraken X60, ensure the cooling port screw has been removed, as that screw will otherwise interfere with the swerve module's Main Plate.



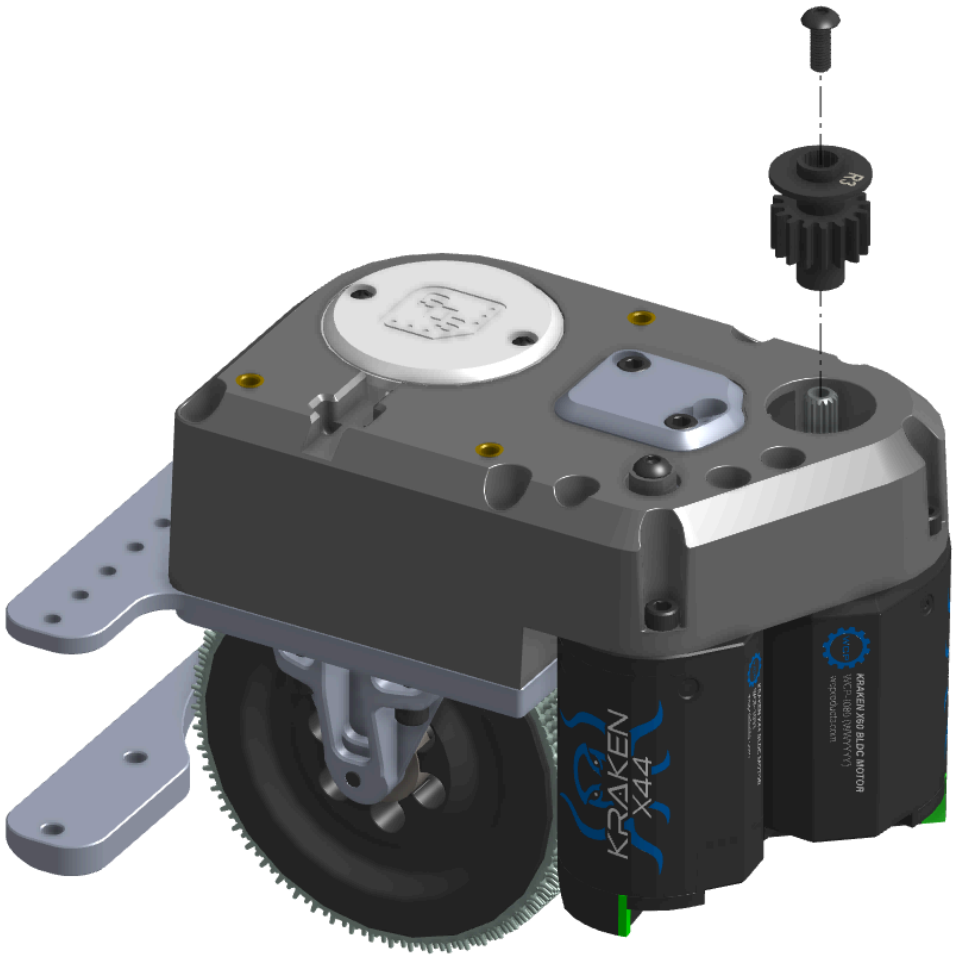
Step 27: Drive Pinion Gear Installation

Place the Drive Pinion Gear onto the motor, and then secure it with a 10-32 x 3/8” button screw (screw drive is T25, not 1/8” hex). Use the following tool on the flats at the top of the gear to prevent it from rotating while tightening the screw:

- Kraken X60: 3/8” open-end wrench
- NEO or NEO Vortex: 1/2” open-end wrench

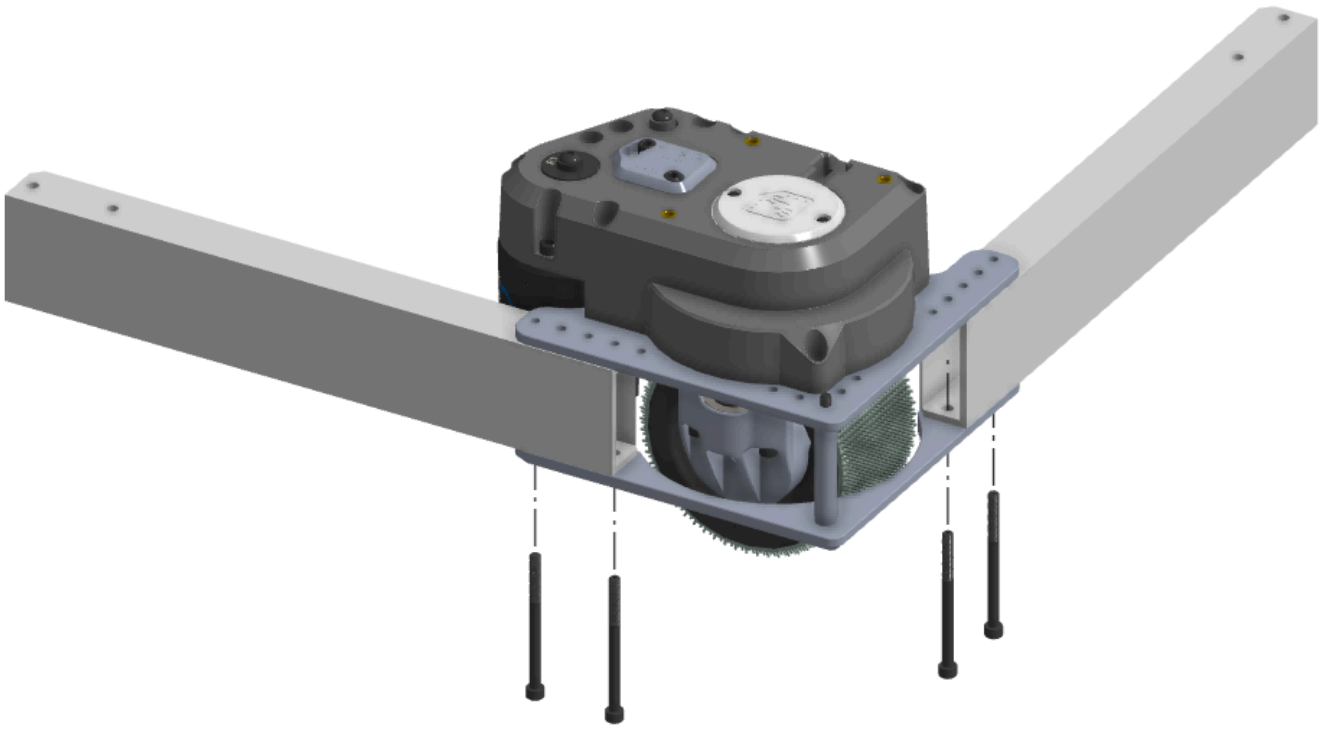
IMPORTANT Ensure that the number marked on the Drive Pinion Gear is the same as the marked pocket the screw in the Shift Plate’s triple-pocket slot is in. Specifically:

- “R1” Drive Pinion Gear and Shift Plate screw pocket “1”
- “R2” Drive Pinion Gear and Shift Plate screw pocket “2”
- “R3” Drive Pinion Gear and Shift Plate screw pocket “3”



Step 28: Swerve Module Installation

The swerve module is typically mounted to 2” x 1” rectangular tubes with four 10-32 x 2.5” socket screws. Refer to the Appendix for additional rectangular tube details.



IMPORTANT Lubricate all gears with Lucas Red “N” Tacky grease regularly with use.

Changing The Drive Gear Ratio

1. Remove the screw securing the Drive Pinion Gear, and then remove the Drive Pinion Gear.
2. Loosen the screw in the Shift Plate's small slot approximately 1/2 turn.
3. Loosen the screw in the triple-pocket of the Shift Plate such that the bottom of the screw head just clears the top of the Shift Plate.
4. Shift the screw in the triple-pocket slot such that the head is over the pocket of the desired ratio (1, 2, or 3).
5. Tighten both screws.
6. Install the Drive Pinion Gear, and then secure it with a 10-32 x 3/8" button screw (screw drive is T25, not 1/8" hex). Use the following tool on the flats at the top of the gear to prevent it from rotating while tightening the screw:
    - Kraken X60: 3/8" open-end wrench
    - NEO or NEO Vortex: 1/2" open-end wrench

IMPORTANT Confirm that the number marked on the Drive Pinion Gear is the same as the marked pocket the screw in the Shift Plate triple-pocket slot is in. Specifically:

- "R1" Drive Pinion Gear and Shift Plate screw pocket "1"
- "R2" Drive Pinion Gear and Shift Plate screw pocket "2"
- "R3" Drive Pinion Gear and Shift Plate screw pocket "3"

Removing The Wheel

1. Remove the 10-32 x 2.5" socket screw going through the wheel.
2. Remove the two 10-32 x 1.25" socket screws from Wheel Mount A, and then remove Wheel Mount A.
3. Remove the wheel (Wheel Mount B does not need to be removed).

Appendix

Frame Tube Dimensions

The swerve module is designed to mount to a frame constructed from 2” x 1” rectangular tube. Each frame member (typically four total) needs the following features at each end:

- Two vertical 0.196” thru-holes for attaching the swerve modules.
- One 3/16” chamfer to clear the wheel.

Pre-drilled tubing that only needs to be cut to length and chamfered is readily available from multiple vendors, including AndyMark. Refer to the following drawing for the frame member size, mounting hole locations, and chamfer locations.

